

Internal Pricing Math — Commercial Roofing Services

Audience: Internal only (owner + estimating staff). Not for customer distribution. **Purpose:** A single sanity-check reference for ballpark pricing across the six commercial roofing services we sell. Backs the public estimator (Task #43) and the customer capabilities sheet (Task #42) so all surfaces use consistent assumptions. **Status:** Draft skeleton. Every dollar figure shown as [PLACEHOLDER: ...] is awaiting real numbers from the owner. Worked examples are intentionally symbolic until those are filled in.

1. Overview

This document captures, for each of our six commercial roofing services:

1. The symbolic pricing formula as a function of roof area.
2. The per-square-foot rate range (low / mid / high) and any fixed/mobilization cost.
3. The minimum project size we will quote.
4. What is included in the rate and what is billed separately.
5. Worked totals at three representative project sizes: **5,000 / 20,000 / 50,000 sq ft**, at low / mid / high.
6. A side-by-side comparison so we can see how the unit cost changes with scale.
7. Edge cases and surcharges that change the math.

The six services covered:

1. TPO / PVC Single-Ply Membrane
 2. Metal Roofing System
 3. Modified Bitumen / Built-Up Roofing (BUR)
 4. Roof Coatings & Restoration
 5. Full Roof Replacement (Tear-Off)
 6. Commercial Roof Repair
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2. Assumptions & Units

- **Area unit:** square feet (sq ft) of roof deck area, measured flat (not surface area of slopes).
- **Currency:** USD, pre-tax.
- **Rounding rule:** All line items computed at full precision, then **total rounded to the nearest \$100**. Effective \$/sq ft figures are reported to two decimals.

- **Tax:** Sales tax is *not* included in any quoted total. Tax handling is applied at invoice time per the customer's exemption status and project jurisdiction.
- **Permits & engineering:** Excluded from the base rate unless explicitly called out. Pass-through at cost + [PLACEHOLDER: permit handling markup %] .
- **Warranty:** Standard manufacturer + workmanship warranty is included in the base rate. Extended / NDL warranties are an add-on.
- **Access assumption:** Single-story, ground-level crane/lift access, no occupied-building constraints, normal working hours. Surcharges below apply when this is not true.
- **Tear-off assumption:** Unless the service is explicitly "Tear-Off" or "Replacement", the base rate assumes the existing roof stays in place (recover or coat). Tear-off is an additive surcharge — see §6.
- **Geography:** North Texas (DFW metro and surrounding counties). All rate ranges below are owner-supplied; this document does not hard-code market numbers.

Symbolic notation

For every service we model total price as:

$$\text{Total}(A) = F + r \cdot A$$

Where:

- A = roof area in sq ft
- F = fixed / mobilization cost (USD) — covers crew mobilization, dumpsters, site setup, project management overhead
- r = per-sq-ft rate (USD/sq ft)

For each service we define three rate points supplied by the owner:

- r_{low} — best-case rate (large, simple, easy-access job)
- r_{high} — worst-case rate (small, complex, restricted-access job)
- r_{mid} — the typical / central case, defined as $r_{\text{mid}} = (r_{\text{low}} + r_{\text{high}}) / 2$ unless the owner supplies a different mid value (e.g. weighted by historical job mix)

Likewise for fixed cost, when the owner gives a range:

- $F_{\text{mid}} = (F_{\text{low}} + F_{\text{high}}) / 2$ unless overridden

For services that tier with scale, we write the rate as $r(A)$:

$$r(A) = \begin{array}{ll} r_{\text{small}} & \text{if } A < A_1 \\ r_{\text{mid}} & \text{if } A_1 \leq A < A_2 \\ r_{\text{large}} & \text{if } A \geq A_2 \end{array}$$

Tier breakpoints A_1 , A_2 are listed per service.

3. Per-Service Pricing Models

Convention in this section: each service shows formula, rate, fixed cost, minimum, inclusions, exclusions, and notes. Every number is owner-supplied and appears as [PLACEHOLDER: ...] . Market "typical" ranges are deliberately not encoded here — quote against owner-supplied numbers only.

3.1 TPO / PVC Single-Ply Membrane

- **Formula:** $\text{Total}(A) = F_{\text{tpo}} + r_{\text{tpo}} \cdot A$
- **Per-sq-ft rate:**
- $r_{\text{tpo},\text{low}} = [\text{PLACEHOLDER: TPO/PVC low \$/sq ft}]$
- $r_{\text{tpo},\text{mid}} = [\text{PLACEHOLDER: TPO/PVC mid \$/sq ft}]$ (default = $(\text{low} + \text{high}) / 2$ if not supplied)
- $r_{\text{tpo},\text{high}} = [\text{PLACEHOLDER: TPO/PVC high \$/sq ft}]$
- **Fixed / mobilization F_{tpo} :** [PLACEHOLDER: TPO mobilization low–mid–high \$]
- **Minimum project size:** [PLACEHOLDER: TPO min sq ft – below this, quote as repair, not system install]
- **Included:** Membrane, insulation board to current code R-value, cover board, fasteners/adhesives, edge metal, standard flashings, manufacturer warranty ([PLACEHOLDER: warranty years]).
- **Excluded:** Tear-off of existing roof, structural deck repair, tapered insulation for drainage correction, skylight/curb replacement, lightning protection, permits.
- **Tier breakpoints (if any):** [PLACEHOLDER: A_1, A_2 for TPO]
- **PVC vs TPO:** PVC pricing uplift over TPO at the same thickness = [PLACEHOLDER: PVC vs TPO uplift %] (for chemical-resistant applications).

3.2 Metal Roofing System

- **Formula:** $\text{Total}(A) = F_{\text{metal}} + r_{\text{metal}} \cdot A$
- **Per-sq-ft rate:**
- $r_{\text{metal},\text{low}} = [\text{PLACEHOLDER: metal low \$/sq ft}]$
- $r_{\text{metal},\text{mid}} = [\text{PLACEHOLDER: metal mid \$/sq ft}]$
- $r_{\text{metal},\text{high}} = [\text{PLACEHOLDER: metal high \$/sq ft}]$
- **Fixed / mobilization F_{metal} :** [PLACEHOLDER: metal mobilization low–mid–high \$] (higher than membrane — crane time, panel staging)
- **Minimum project size:** [PLACEHOLDER: metal min sq ft]
- **Included:** Standing-seam panels, clips, underlayment, trim/flashings, snow guards (if specified), manufacturer finish warranty.
- **Excluded:** Tear-off, deck or purlin repair, gutter/downspout replacement, insulation upgrade beyond existing R-value, custom color premium.

- **Notes:** Panel length, gauge, and finish (Galvalume vs Kynar) materially change `r_metal`. Long single-run panels (`> [PLACEHOLDER: panel length ft]`) trigger crane / specialty transport surcharge.

3.3 Modified Bitumen / Built-Up Roofing (BUR)

- **Formula:** $\text{Total}(A) = F_{\text{bur}} + r_{\text{bur}} \cdot A$
- **Per-sq-ft rate:**
 - `r_bur,low` = `[PLACEHOLDER: mod bit/BUR low $/sq ft]`
 - `r_bur,mid` = `[PLACEHOLDER: mod bit/BUR mid $/sq ft]`
 - `r_bur,high` = `[PLACEHOLDER: mod bit/BUR high $/sq ft]`
- **Fixed / mobilization `F_bur`:** `[PLACEHOLDER: BUR mobilization low-mid-high $]`
- **Minimum project size:** `[PLACEHOLDER: BUR min sq ft]`
- **Included:** Base + cap sheet, insulation, cover board, granulated cap, flashings.
- **Excluded:** Hot-kettle setup surcharge for traditional BUR (asphalt), tear-off, deck repair.
- **Notes:** Torch-down installations require fire-watch coverage; base rate reflects standard SBS self-adhered unless otherwise noted.

3.4 Roof Coatings & Restoration

- **Formula:** $\text{Total}(A) = F_{\text{coat}} + r_{\text{coat}} \cdot A$
- **Per-sq-ft rate:**
 - `r_coat,low` = `[PLACEHOLDER: coating low $/sq ft]`
 - `r_coat,mid` = `[PLACEHOLDER: coating mid $/sq ft]`
 - `r_coat,high` = `[PLACEHOLDER: coating high $/sq ft]`
- **Fixed / mobilization `F_coat`:** `[PLACEHOLDER: coating mobilization low-mid-high $]`
- **Minimum project size:** `[PLACEHOLDER: coating min sq ft]`
- **Included:** Power-wash, primer (if required by substrate), 1-2 coats of silicone/acrylic at spec mils, embedded fabric at seams and penetrations, manufacturer restoration warranty (`[PLACEHOLDER: warranty years]`).
- **Excluded:** Substrate repair beyond `[PLACEHOLDER: included repair sq ft]`, ponding-water correction, removal of failed prior coating.
- **Eligibility:** Coating is only sold when the existing roof passes a restorability check — substrate must be $\geq$ `[PLACEHOLDER: min remaining life years]` of remaining life, $<$ `[PLACEHOLDER: max saturated insulation %]` saturated insulation, no widespread membrane failure. If it fails, we quote replacement or recover instead.

3.5 Full Roof Replacement (Tear-Off)

- **Formula:** $\text{Total}(A) = F_{\text{repl}} + (r_{\text{system}} + r_{\text{tearoff}} + r_{\text{disposal}}) \cdot A$
- **Per-sq-ft rate components:**
 - `r_system` = whichever new system rate from §3.1 – §3.3 the customer selects (low / mid / high accordingly)

- $r_{\text{tearoff,low / mid / high}} = [\text{PLACEHOLDER: tear-off labor low / mid / high \$/sq ft}]$
- $r_{\text{disposal,low / mid / high}} = [\text{PLACEHOLDER: disposal low / mid / high \$/sq ft}]$ (function of dump fees and ply count)
- **Fixed / mobilization F_{repl}** : $[\text{PLACEHOLDER: replacement mobilization low-mid-high \$}]$ (larger than non-tear-off — more dumpsters, more days)
- **Minimum project size:** $[\text{PLACEHOLDER: replacement min sq ft}]$
- **Included:** Removal of all existing roofing plies down to deck, deck inspection, up to $[\text{PLACEHOLDER: included deck repair sq ft}]$ of deck repair at no charge, new system per chosen spec, all flashings.
- **Excluded:** Deck repair beyond the included allowance (billed at $[\text{PLACEHOLDER: deck repair \$/sq ft}]$), asbestos abatement, structural reinforcement.
- **Notes:** When existing roof has > 2 plies or is wet, r_{tearoff} and r_{disposal} move to the upper end of the range.

3.6 Commercial Roof Repair

- **Formula:** $\text{Total} = \max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair}} \cdot A_{\text{affected}})$
- **Per-sq-ft rate r_{repair}** (per sq ft of *affected* area, not roof area):
- $r_{\text{repair,low}} = [\text{PLACEHOLDER: repair low \$/sq ft of affected area}]$
- $r_{\text{repair,mid}} = [\text{PLACEHOLDER: repair mid \$/sq ft of affected area}]$
- $r_{\text{repair,high}} = [\text{PLACEHOLDER: repair high \$/sq ft of affected area}]$
- **Trip / mobilization F_{repair}** : $[\text{PLACEHOLDER: repair trip charge \$}]$
- **Minimum invoice MinRepair** : $[\text{PLACEHOLDER: minimum repair invoice \$}]$
- **Included:** Diagnosis, patch of identified leak(s), photo documentation, workmanship warranty on patched area ($[\text{PLACEHOLDER: warranty days}]$).
- **Excluded:** Investigation of unrelated leaks discovered on site (quoted separately), interior water-damage repair, anything outside the patched area.
- **Notes:** Repair is the only service where A in the formula is *affected area*, not total roof area. Do **not** quote repair against full-roof sq ft. See §4.6 for how the 5k/20k/50k framing applies to repair (it scales with affected area as a fraction of roof area, not roof area itself).

4. Worked Examples

All examples use the symbolic forms above. Substitute owner-supplied F and r when they are provided; totals will resolve to numeric values. For each service we show **low / mid / high** at each representative size, with the explicit substitution step. Mid uses r_{mid} and F_{mid} as defined in §2 ($= (\text{low} + \text{high})/2$ unless overridden).

For each service:

$$\begin{aligned} \text{Total_low}(A) &= F_{\text{low}} + r_{\text{low}} \cdot A \\ \text{Total_mid}(A) &= F_{\text{mid}} + r_{\text{mid}} \cdot A \\ \text{Total_high}(A) &= F_{\text{high}} + r_{\text{high}} \cdot A \end{aligned}$$

All totals rounded to the nearest \$100 (see §2).

4.1 TPO / PVC Single-Ply Membrane

Area A	Tier	Substitution	Total (→ \$100)
5,000	low	$F_{\text{tpo,low}} + r_{\text{tpo,low}} \cdot 5,000$	[PLACEHOLDER]
5,000	mid	$F_{\text{tpo,mid}} + r_{\text{tpo,mid}} \cdot 5,000$	[PLACEHOLDER]
5,000	high	$F_{\text{tpo,high}} + r_{\text{tpo,high}} \cdot 5,000$	[PLACEHOLDER]
20,000	low	$F_{\text{tpo,low}} + r_{\text{tpo,low}} \cdot 20,000$	[PLACEHOLDER]
20,000	mid	$F_{\text{tpo,mid}} + r_{\text{tpo,mid}} \cdot 20,000$	[PLACEHOLDER]
20,000	high	$F_{\text{tpo,high}} + r_{\text{tpo,high}} \cdot 20,000$	[PLACEHOLDER]
50,000	low	$F_{\text{tpo,low}} + r_{\text{tpo,low}} \cdot 50,000$	[PLACEHOLDER]
50,000	mid	$F_{\text{tpo,mid}} + r_{\text{tpo,mid}} \cdot 50,000$	[PLACEHOLDER]
50,000	high	$F_{\text{tpo,high}} + r_{\text{tpo,high}} \cdot 50,000$	[PLACEHOLDER]

4.2 Metal Roofing System

Area A	Tier	Substitution	Total (→ \$100)
5,000	low	$F_{\text{metal,low}} + r_{\text{metal,low}} \cdot 5,000$	[PLACEHOLDER]
5,000	mid	$F_{\text{metal,mid}} + r_{\text{metal,mid}} \cdot 5,000$	[PLACEHOLDER]
5,000	high	$F_{\text{metal,high}} + r_{\text{metal,high}} \cdot 5,000$	[PLACEHOLDER]
20,000	low	$F_{\text{metal,low}} + r_{\text{metal,low}} \cdot 20,000$	[PLACEHOLDER]
20,000	mid	$F_{\text{metal,mid}} + r_{\text{metal,mid}} \cdot 20,000$	[PLACEHOLDER]
20,000	high	$F_{\text{metal,high}} + r_{\text{metal,high}} \cdot 20,000$	[PLACEHOLDER]
50,000	low	$F_{\text{metal,low}} + r_{\text{metal,low}} \cdot 50,000$	[PLACEHOLDER]
50,000	mid	$F_{\text{metal,mid}} + r_{\text{metal,mid}} \cdot 50,000$	[PLACEHOLDER]
50,000	high	$F_{\text{metal,high}} + r_{\text{metal,high}} \cdot 50,000$	[PLACEHOLDER]

4.3 Modified Bitumen / BUR

Area A	Tier	Substitution	Total (→ \$100)
5,000	low	$F_{bur,low} + r_{bur,low} \cdot 5,000$	[PLACEHOLDER]
5,000	mid	$F_{bur,mid} + r_{bur,mid} \cdot 5,000$	[PLACEHOLDER]
5,000	high	$F_{bur,high} + r_{bur,high} \cdot 5,000$	[PLACEHOLDER]
20,000	low	$F_{bur,low} + r_{bur,low} \cdot 20,000$	[PLACEHOLDER]
20,000	mid	$F_{bur,mid} + r_{bur,mid} \cdot 20,000$	[PLACEHOLDER]
20,000	high	$F_{bur,high} + r_{bur,high} \cdot 20,000$	[PLACEHOLDER]
50,000	low	$F_{bur,low} + r_{bur,low} \cdot 50,000$	[PLACEHOLDER]
50,000	mid	$F_{bur,mid} + r_{bur,mid} \cdot 50,000$	[PLACEHOLDER]
50,000	high	$F_{bur,high} + r_{bur,high} \cdot 50,000$	[PLACEHOLDER]

4.4 Roof Coatings & Restoration

Area A	Tier	Substitution	Total (→ \$100)
5,000	low	$F_{coat,low} + r_{coat,low} \cdot 5,000$	[PLACEHOLDER]
5,000	mid	$F_{coat,mid} + r_{coat,mid} \cdot 5,000$	[PLACEHOLDER]
5,000	high	$F_{coat,high} + r_{coat,high} \cdot 5,000$	[PLACEHOLDER]
20,000	low	$F_{coat,low} + r_{coat,low} \cdot 20,000$	[PLACEHOLDER]
20,000	mid	$F_{coat,mid} + r_{coat,mid} \cdot 20,000$	[PLACEHOLDER]
20,000	high	$F_{coat,high} + r_{coat,high} \cdot 20,000$	[PLACEHOLDER]
50,000	low	$F_{coat,low} + r_{coat,low} \cdot 50,000$	[PLACEHOLDER]
50,000	mid	$F_{coat,mid} + r_{coat,mid} \cdot 50,000$	[PLACEHOLDER]
50,000	high	$F_{coat,high} + r_{coat,high} \cdot 50,000$	[PLACEHOLDER]

4.5 Full Roof Replacement (Tear-Off + TPO assumed)

Assumes $r_{system} = r_{tpo}$ for this worked example. Swap in metal or BUR rates as needed.

Let $r_{repl,tier} = r_{tpo,tier} + r_{tearoff,tier} + r_{disposal,tier}$ for $tier \in \{low, mid, high\}$.

Area A	Tier	Substitution	Total (→ \$100)
5,000	low	$F_{\text{repl,low}} + r_{\text{repl,low}} \cdot 5,000$	[PLACEHOLDER]
5,000	mid	$F_{\text{repl,mid}} + r_{\text{repl,mid}} \cdot 5,000$	[PLACEHOLDER]
5,000	high	$F_{\text{repl,high}} + r_{\text{repl,high}} \cdot 5,000$	[PLACEHOLDER]
20,000	low	$F_{\text{repl,low}} + r_{\text{repl,low}} \cdot 20,000$	[PLACEHOLDER]
20,000	mid	$F_{\text{repl,mid}} + r_{\text{repl,mid}} \cdot 20,000$	[PLACEHOLDER]
20,000	high	$F_{\text{repl,high}} + r_{\text{repl,high}} \cdot 20,000$	[PLACEHOLDER]
50,000	low	$F_{\text{repl,low}} + r_{\text{repl,low}} \cdot 50,000$	[PLACEHOLDER]
50,000	mid	$F_{\text{repl,mid}} + r_{\text{repl,mid}} \cdot 50,000$	[PLACEHOLDER]
50,000	high	$F_{\text{repl,high}} + r_{\text{repl,high}} \cdot 50,000$	[PLACEHOLDER]

4.6 Commercial Roof Repair — mapped onto the 5k / 20k / 50k framing

Repair does **not** scale with total roof area — it scales with the *affected area* on that roof. To keep this section comparable to §4.1 – §4.5, we map each representative roof size to an affected-area fraction φ (the share of the roof that needs patching), then compute $A_{\text{affected}} = \varphi \cdot A$.

The owner sets one fraction per tier; the same fractions are used across all roof sizes:

- φ_{low} = [PLACEHOLDER: affected-area fraction, low – small leak / one penetration]
- φ_{mid} = [PLACEHOLDER: affected-area fraction, mid – typical service call]
- φ_{high} = [PLACEHOLDER: affected-area fraction, high – widespread punctures, borderline replacement]

Then for each roof size A and tier:

$$A_{\text{affected}} = \varphi_{\text{tier}} \cdot A$$

$$\text{Total}_{\text{tier}}(A) = \max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair,tier}} \cdot A_{\text{affected}})$$

Roof size A	Tier	A_affected = φ · A	Substitution	Total (→ \$100)
5,000	low	$\varphi_{\text{low}} \cdot 5,000$	$\max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair,low}} \cdot \varphi_{\text{low}} \cdot 5,000)$	[PLACEHOLDER]
5,000	mid	$\varphi_{\text{mid}} \cdot 5,000$	$\max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair,mid}} \cdot \varphi_{\text{mid}} \cdot 5,000)$	[PLACEHOLDER]
5,000	high	$\varphi_{\text{high}} \cdot 5,000$	$\max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair,high}} \cdot \varphi_{\text{high}} \cdot 5,000)$	[PLACEHOLDER]
20,000	low	$\varphi_{\text{low}} \cdot 20,000$	$\max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair,low}} \cdot \varphi_{\text{low}} \cdot 20,000)$	[PLACEHOLDER]
20,000	mid	$\varphi_{\text{mid}} \cdot 20,000$	$\max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair,mid}} \cdot \varphi_{\text{mid}} \cdot 20,000)$	[PLACEHOLDER]
20,000	high	$\varphi_{\text{high}} \cdot 20,000$	$\max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair,high}} \cdot \varphi_{\text{high}} \cdot 20,000)$	[PLACEHOLDER]
50,000	low	$\varphi_{\text{low}} \cdot 50,000$	$\max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair,low}} \cdot \varphi_{\text{low}} \cdot 50,000)$	[PLACEHOLDER]
50,000	mid	$\varphi_{\text{mid}} \cdot 50,000$	$\max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair,mid}} \cdot \varphi_{\text{mid}} \cdot 50,000)$	[PLACEHOLDER]
50,000	high	$\varphi_{\text{high}} \cdot 50,000$	$\max(\text{MinRepair}, F_{\text{repair}} + r_{\text{repair,high}} \cdot \varphi_{\text{high}} \cdot 50,000)$	[PLACEHOLDER]

Watch for `MinRepair` binding on small roofs at low φ — for a small leak on a small building the floor often dominates. If the unbounded computation falls below `MinRepair`, the floor is the price.

5. Side-by-Side Comparison

Two views: **total cost** at each example size (low / mid / high), and **effective \$/sq ft** (= mid total ÷ area) so we can see how unit cost compresses with scale because the fixed `F` is amortized over more area.

5.1 Total cost by service and size — low / mid / high

Service	5,000 sq ft (low / mid / high)	20,000 sq ft (low / mid / high)	50,000 sq ft (low / mid / high)
TPO / PVC	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]
Metal	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]
Mod Bit / BUR	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]
Coatings / Restoration	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]
Full Replacement (w/ TPO)	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]
Repair (mapped, see §4.6)	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]	[PLACEHOLDER] / [PLACEHOLDER] / [PLACEHOLDER]

5.2 Effective \$/sq ft by service and size (using mid)

Effective \$/sq ft = $\text{Total_mid}(A) / A$. Watch how it falls as A grows — that's F_{mid}/A shrinking.

Service	5,000 sq ft	20,000 sq ft	50,000 sq ft
TPO / PVC	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
Metal	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
Mod Bit / BUR	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
Coatings / Restoration	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
Full Replacement (w/ TPO)	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
Repair (mapped)	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]

5.3 What the table should tell us once filled in

- Coating should be the lowest \$/sq ft at every size — it's a restoration, not a new system.
- Metal should be the highest \$/sq ft at every size.

- Full Replacement should always be strictly greater than TPO alone at the same size — by roughly $(r_{\text{tearoff}} + r_{\text{disposal}}) \cdot A + (F_{\text{repl}} - F_{\text{tpo}})$.
- The 5,000 sq ft column should show a noticeably higher \$/sq ft than 50,000 for every service. If it doesn't, our F is probably set too low.
- Repair's effective \$/sq ft (mapped against full roof area) should fall sharply as A grows when ϕ is held constant — because MinRepair and F_{repair} are amortized over a larger denominator.

6. Edge Cases, Surcharges & Minimums

These adjust the base formula. Apply them after the base $\text{Total}(A)$ is computed.

6.1 Repair minimum invoice

- Repair total is bounded below by MinRepair (see §3.6). If $F_{\text{repair}} + r_{\text{repair}} \cdot A_{\text{affected}} < \text{MinRepair}$, bill MinRepair .
- Do **not** waive MinRepair for an existing customer without owner approval — it represents real truck-roll cost.

6.2 Coating restorability gate

Before quoting coating, the substrate must pass all of:

- Remaining membrane life $\geq$ [PLACEHOLDER: min remaining life years].
- Saturated insulation $<$ [PLACEHOLDER: max saturated insulation %] of roof area (confirm with infrared or core cuts).
- No active structural deck deflection or rot.
- Prior coating, if any, is compatible or fully removed (removal billed separately).

If any check fails, quote replacement (§3.5) or recover instead.

6.3 Tear-off vs recover decision

- Recover (install new system over existing) is permitted when the existing roof has ≤ 1 ply and is dry — saves the customer roughly $(r_{\text{tearoff}} + r_{\text{disposal}}) \cdot A$.
- Tear-off is required when there are already 2 plies (code), or any saturation, or substrate failure.
- Document the decision in the estimate notes; it is the single biggest swing factor on a replacement quote.

6.4 Access and height surcharges

Add to the base total when the access assumption in §2 does not hold:

- **Multi-story (≥ 2 stories above grade):** + [PLACEHOLDER: multi-story uplift % of base] .
- **Crane required (no truck-mounted lift access):** + [PLACEHOLDER: crane day rate \$] · [days] .
- **After-hours / occupied-building (nights, weekends, hospitals, schools):** + [PLACEHOLDER: after-hours uplift % of labor] .
- **Restricted lay-down area:** + [PLACEHOLDER: staging surcharge \$] .

6.5 Tax handling

- Quoted totals are pre-tax.
- For tax-exempt customers, obtain the exemption certificate **before** invoicing.
- For taxable customers, sales tax is computed on materials (and on labor where the jurisdiction taxes it) at invoice time.

6.6 Flagging missing inputs

Any line that still reads [PLACEHOLDER: ...] when this document is used to back a customer-facing number is a blocker. Do not quote against placeholders — escalate to the owner to fill in the real range, then re-run the math.

7. Conclusions / How to Use

1. **Ballpark a job:** Pick the service, plug A into $\text{Total}(A) = F + r \cdot A$, use low / mid / high to give a range and a central estimate, round each to the nearest \$100.
2. **Sanity-check a vendor estimator number:** Drop the estimator's total into the \$5 comparison table at the customer's roof size. If it falls outside the low-high band by more than [PLACEHOLDER: tolerance %], dig into which surcharge or exclusion explains the gap.
3. **Train staff:** Walk new estimators through §3 and §6 first — the formula is easy, the exclusions and surcharges are where mistakes happen.
4. **Feed the public estimator (Task #43):** Public surface should show a range only (anchored on mid, bounded by low/high), never a single number, and never go below the minimums in §3 or §6.1.
5. **Keep this doc current:** When the owner updates a rate, edit the corresponding [PLACEHOLDER] in §3, then re-render the PDF (see docs/pricing/README.md).